

Invariant Preservation under Role Transposition

Theorem 8.1 of the Directive-Based Products Framework

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1 Setting

We work within the Directive-Based Products (DBP) framework. The objects of study are *well-posed triples* $\tau = (D, S, P)$ satisfying a coupling relation $D \oplus S = P$, where $\oplus$ is a binary operation satisfying five standing properties (existence of forward map, existence of both inverses, measurability of invariants, and continuity). The specific properties are not needed for this proof; what matters is that $\oplus$ determines P from (D, S) and admits recovery operations $\ominus_D$ and $\ominus_S$ such that $S = P \ominus_D D$ and $D = P \ominus_S S$.

Definition 1 (Role transposition). The *transposition operator* $\mathcal{T}$ is the action of the symmetric group S_3 on well-posed triples. Each $\sigma \in S_3$ reassigns which value occupies the Directive, Substrate, and Product roles while preserving the coupling relation $\oplus$. The six elements of S_3 correspond to the six ways of reading the coupling equation. The precise form of each transposition depends on the coupling; what matters is that $\mathcal{T}_\sigma$ permutes the roles while maintaining the constraint $D \oplus S = P$.

Definition 2 (Coupling kernel). The *coupling kernel* $K(\tau)$ of a well-posed triple $\tau = (D, S, P)$ is the tuple

$$K = (K_{\text{spec}}, K_{\text{hol}}, \kappa),$$

where K_{spec} is the spectral data of $\oplus$ at the operating point (including the Jacobian $J_\oplus = (\partial P/\partial D, \partial P/\partial S)$ and higher-order derivative data), K_{hol} is the holonomy residue from the full role cycle $D \rightarrow S \rightarrow P \rightarrow D$, and $\kappa \in [0, 1]$ is the intrinsic nonlinearity parameter. K captures the local coupling-relational content of the triple: how D , S , and P are related through $\oplus$ and its derivatives at the operating point.

Definition 3 (DBP invariant). A function φ on well-posed triples is a *DBP invariant* if $\varphi(\tau) = \varphi(\mathcal{T}_\sigma(\tau))$ for all $\sigma \in S_3$.

Definition 4 (Factors through K). A function φ *factors through* K if there exists a function ψ such that $\varphi(\tau) = \psi(K(\tau))$ for all well-posed τ .

Definition 5 (Coupling-structural). A function φ on well-posed triples is *coupling-structural* if it cannot be expressed as a function of the unordered multiset $\{D, S, P\}$ alone. That is, there is no function g such that

$$\varphi(D, S, P) = g(\{D, S, P\})$$

for all well-posed triples (D, S, P) .

Remark 1 (Equivalent external formulation). Definition 5 is equivalent to: φ is coupling-structural if and only if there exist two couplings $\oplus_1, \oplus_2$ and values a, b with $a \oplus_1 b = a \oplus_2 b$ such that $\varphi(a, b, a \oplus_1 b; \oplus_1) \neq \varphi(a, b, a \oplus_2 b; \oplus_2)$.

The equivalence is immediate: a function of the multiset $\{D, S, P\}$ does not reference $\oplus$ and therefore assigns the same value regardless of which coupling produced the triple. Conversely, a function that is not expressible from the multiset alone must depend on coupling structure, which different couplings can realise differently at the same values.

Definition 6 (Value-symmetric). A function φ on triples is *value-symmetric* if $\varphi(D, S, P) = \varphi(\pi(D, S, P))$ for every permutation π of the three values, regardless of coupling. Equivalently, φ depends only on the unordered multiset $\{D, S, P\}$, or equivalently, φ factors through the elementary symmetric polynomials $e_1 = D + S + P$, $e_2 = DS + DP + SP$, $e_3 = DSP$.

Remark 2. The coupling-structural and value-symmetric classes are complementary: a function is coupling-structural if and only if it is not value-symmetric.

2 Main result

Theorem 1 (Invariant Preservation). *Let φ be a function on well-posed DBP triples.*

- (i) *K is $\mathcal{T}$ -invariant: $K(\mathcal{T}_\sigma(\tau)) = K(\tau)$ for all $\sigma \in S_3$.*
- (ii) *If φ factors through K , then φ is a DBP invariant.*
- (iii) *If φ is coupling-structural and a DBP invariant, then φ factors through K .*
- (iv) *The DBP invariants that do not factor through K are precisely the value-symmetric functions.*

Proof. Part (i): K is $\mathcal{T}$ -invariant.

K is computed from the coupling relation $\oplus$ and its derivatives at the operating point. A role transposition $\mathcal{T}_\sigma$ re-expresses $\oplus$ in a different frame—it relabels which value is called D , which S , which P —but does not change the coupling itself. The Jacobian $J_\oplus$, the holonomy K_{hol} , and the nonlinearity κ are all defined in terms of $\oplus$ and its inverses, not in terms of which value occupies which role. Therefore $K(\mathcal{T}_\sigma(\tau)) = K(\tau)$.

Part (ii): Forward direction.

Assume $\varphi = \psi \circ K$. Since K is $\mathcal{T}$ -invariant by part (i):

$$\varphi(\mathcal{T}_\sigma(\tau)) = \psi(K(\mathcal{T}_\sigma(\tau))) = \psi(K(\tau)) = \varphi(\tau). \quad \square$$

Part (iii): Reverse direction (maximality of K).

We prove the contrapositive: if φ is coupling-structural and does *not* factor through K , then φ is not a DBP invariant.

At any operating point, a function on well-posed triples can depend on exactly two types of information:

- (a) **Coupling data:** the local structure of $\oplus$ at the operating point—the Jacobian, holonomy, nonlinearity, and higher-order derivatives. This is precisely what K captures.

- (b) **Role-labelled values:** which specific number sits in the D -slot, which in the S -slot, which in the P -slot. This is the information that $\mathcal{T}$ permutes.

These two types exhaust the information content of a well-posed triple at a point: the coupling data determines how D, S, P are *related* through $\oplus$, and the role-labelled values determine which specific values *instantiate* those roles. Together they determine the triple completely.

Now suppose φ is coupling-structural and does not factor through K . We have a dichotomy:

Subcase (a): φ depends only on coupling data. Then φ is a function of K alone, contradicting the assumption that φ does not factor through K .

Subcase (b): φ depends (also) on role-labelled values. Then there exist two triples $\tau = (D, S, P)$ and $\tau' = (D', S', P')$ with the same coupling data $K(\tau) = K(\tau')$ and the same unordered values $\{D, S, P\} = \{D', S', P'\}$, but with different role assignments (a different value in the D -slot, say), such that $\varphi(\tau) \neq \varphi(\tau')$.

Since τ and τ' have the same unordered values and satisfy the same coupling, $\tau' = \mathcal{T}_\sigma(\tau)$ for some $\sigma \in S_3$. (This follows directly from Definition 1: $\mathcal{T}_\sigma$ acts by reassigning which value occupies each role while preserving the coupling, which is exactly the operation relating τ and τ' .)

Therefore $\varphi(\tau) \neq \varphi(\mathcal{T}_\sigma(\tau))$, which means φ is not a DBP invariant.

In both subcases, the assumption that φ is coupling-structural and does not factor through K leads to a contradiction with $\mathcal{T}$ -invariance. By contrapositive: if φ is coupling-structural and $\mathcal{T}$ -invariant, then φ factors through K .

Part (iv): Residual invariants.

We show that the $\mathcal{T}$ -invariant functions not factoring through K are precisely the value-symmetric functions.

($\subseteq$) Let φ be $\mathcal{T}$ -invariant but not coupling-structural. By Definition 5, φ is expressible as a function of the multiset $\{D, S, P\}$, i.e. φ is value-symmetric.

($\supseteq$) Let φ be value-symmetric. Then φ depends only on the multiset $\{D, S, P\}$, which is unchanged by any permutation of roles. Therefore φ is $\mathcal{T}$ -invariant.

Corollary 2 (Exhaustive decomposition). *Every $\mathcal{T}$ -invariant function φ decomposes uniquely as*

$$\varphi = \varphi_K + \varphi_{\text{sym}},$$

where φ_K factors through K (coupling-structural component) and φ_{sym} factors through the elementary symmetric polynomials e_1, e_2, e_3 (value-symmetric component). There is no third category.

Proof. Existence. Define $\varphi_{\text{sym}}(D, S, P)$ as the average of φ over all 6 permutations of (D, S, P) as values (ignoring the coupling):

$$\varphi_{\text{sym}}(D, S, P) = \frac{1}{6} \sum_{\pi \in S_3} \varphi(\pi(D), \pi(S), \pi(P)).$$

By construction, φ_{sym} is value-symmetric and hence factors through e_1, e_2, e_3 .

Define $\varphi_K = \varphi - \varphi_{\text{sym}}$. Since φ is $\mathcal{T}$ -invariant and φ_{sym} is $\mathcal{T}$ -invariant (being value-symmetric), their difference φ_K is $\mathcal{T}$ -invariant. By construction, φ_K has zero value-symmetric component (its symmetrisation vanishes), so it is coupling-structural. By part (iii) of Theorem 1, φ_K factors through K .

Uniqueness. Suppose $\varphi = \varphi_K + \varphi_{\text{sym}} = \varphi'_K + \varphi'_{\text{sym}}$. Then $r = \varphi_K - \varphi'_K = \varphi'_{\text{sym}} - \varphi_{\text{sym}}$. The left side is coupling-structural (difference of functions factoring through K , with vanishing symmetrisation). The right side is value-symmetric. A function that is simultaneously coupling-structural and value-symmetric is a contradiction in terms (it would need to be both expressible and not expressible from the multiset alone). Therefore $r = 0$. $\square$

Corollary 3 (Nonlinear domains). *For couplings with $\kappa > 0$ at generic operating points, the Jacobian $J_{\oplus}$ varies across parameter space. Therefore K generically separates S_3 -orbits, and the coupling-structural qualification in part (iii) is unnecessary: every $\mathcal{T}$ -invariant function factors through K .*

Proof. If $\kappa > 0$ generically, then $J_{\oplus}$ is non-constant. Two triples with $K(\tau_1) = K(\tau_2)$ must share the same Jacobian and hence (for analytic $\oplus$) the same operating point up to the discrete ambiguity of S_3 role permutation. Therefore $\tau_2 = \mathcal{T}_{\sigma}(\tau_1)$ for some σ , and K is a complete invariant. Value-symmetric functions become functions of K as well, since K determines the values up to role permutation, which the elementary symmetric polynomials do not distinguish. $\square$

3 Scope

Theorem 1 is a *local* result. K captures coupling information up to the order of its components: K_{spec} encodes first-order structure (the Jacobian), K_{hol} encodes second-order structure (curvature of the coupling surface), and κ constrains global character.

The dichotomy argument in part (iii) does not depend on the specific contents of K —only on the partition between coupling data and role-labelled values. Therefore the theorem extends to any augmentation of K with higher-order derivative data. At each augmentation level, K captures all coupling-structural invariants of that order.

Global and topological invariants of the coupling surface (properties that cannot be detected from any finite-order jet at a single point) are not addressed. The theorem concerns local coupling structure at individual operating points.

4 Discussion

The theorem establishes a clean separation. The information content of a $\mathcal{T}$ -invariant quantity comes from exactly one of two sources: the coupling structure (captured by K) or the value configuration (captured by the elementary symmetric polynomials). There is no mixing.

The coupling-structural invariants are the diagnostically relevant ones—they encode *how* D , S , P are related through $\oplus$. The value-symmetric invariants are diagnostically inert—they carry no information about the coupling and would be invariant under any relabeling, not just coupling-preserving transpositions.

For nonlinear couplings ($\kappa > 0$), which include all diagnostically interesting domains tested to date, the distinction collapses: K is a complete invariant and captures everything.